

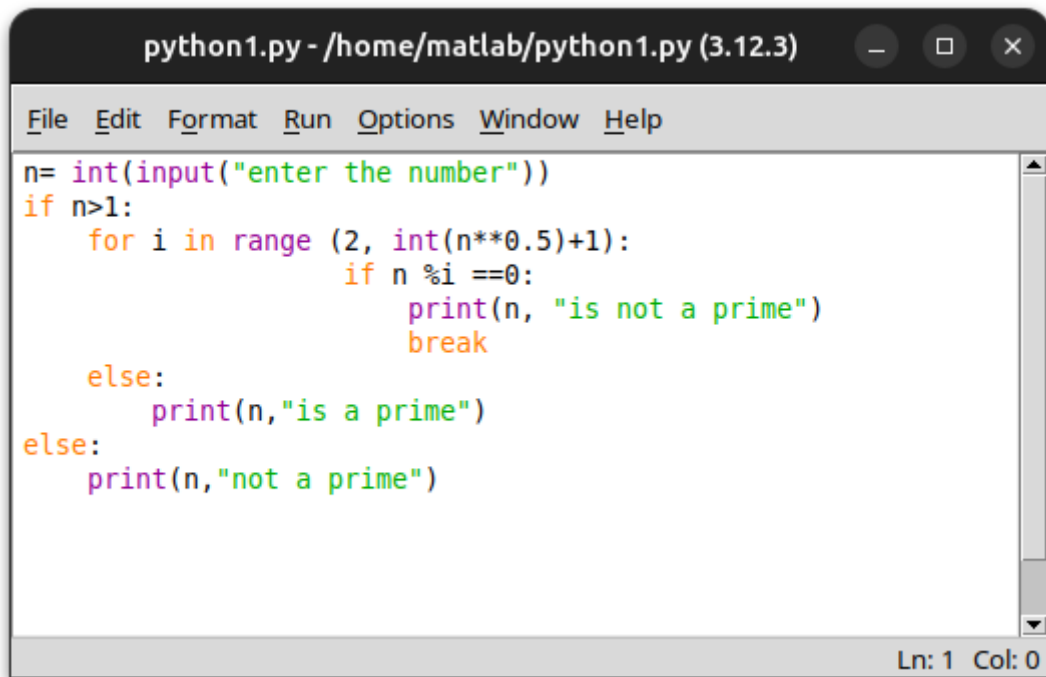
Subject: Exploratory Data analysis

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Experiment:
Python Programs

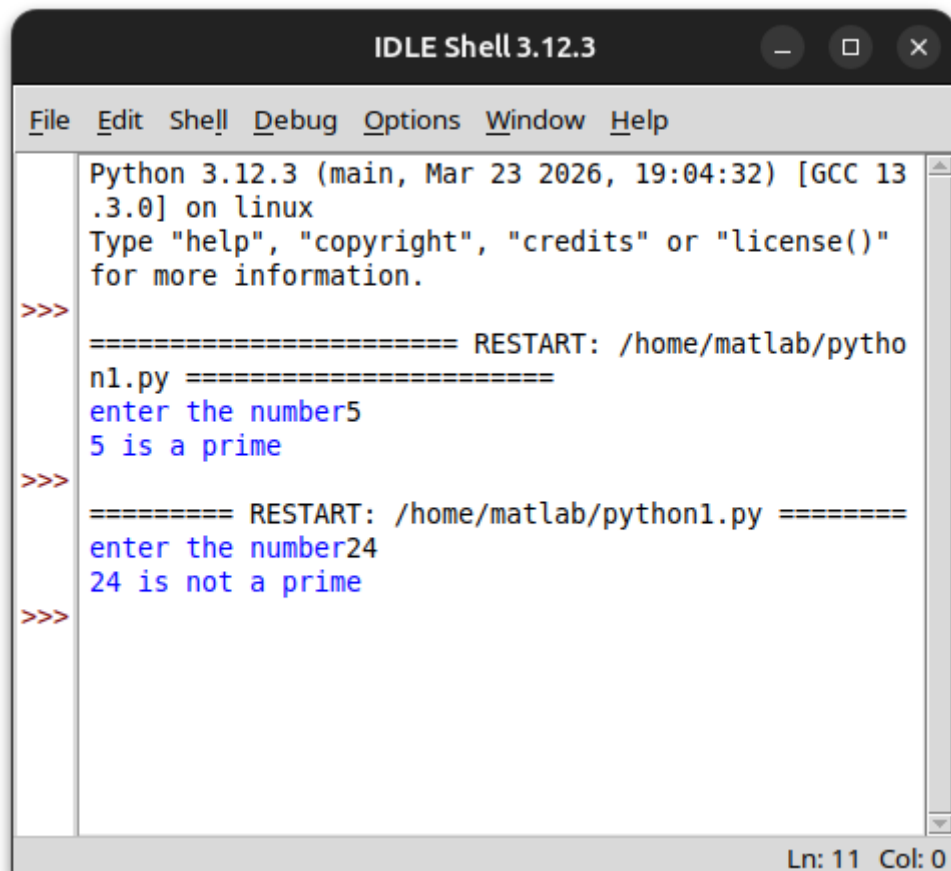
1.



The screenshot shows a Python script editor window. The title bar reads "python1.py - /home/matlab/python1.py (3.12.3)". The menu bar includes "File", "Edit", "Format", "Run", "Options", "Window", and "Help". The code is as follows:

```
n= int(input("enter the number"))
if n>1:
    for i in range (2, int(n**0.5)+1):
        if n %i ==0:
            print(n, "is not a prime")
            break
    else:
        print(n,"is a prime")
else:
    print(n,"not a prime")
```

The status bar at the bottom right indicates "Ln: 1 Col: 0".

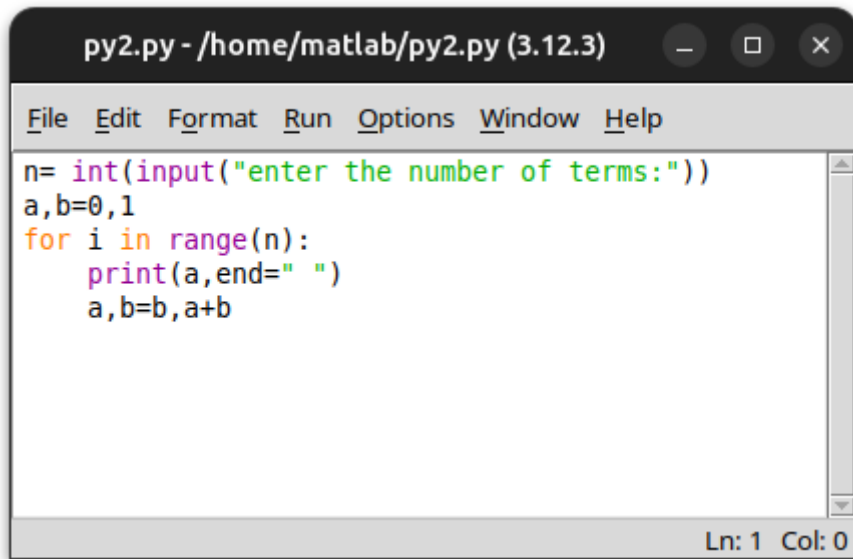


The screenshot shows the IDLE Shell 3.12.3 window. The title bar reads "IDLE Shell 3.12.3". The menu bar includes "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The shell output is as follows:

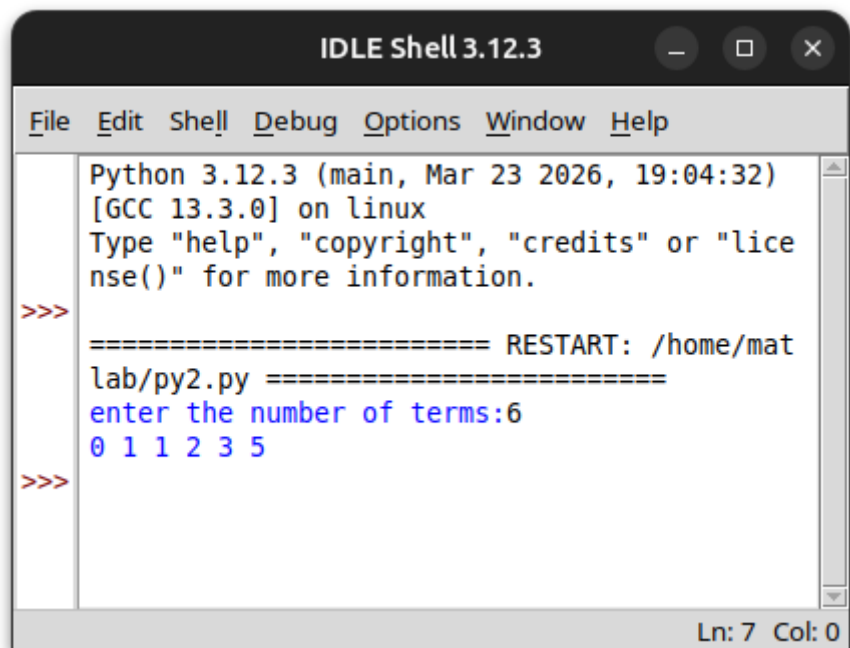
```
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13
.3.0] on linux
Type "help", "copyright", "credits" or "license()"
for more information.
>>>
===== RESTART: /home/matlab/pytho
n1.py =====
enter the number5
5 is a prime
>>>
===== RESTART: /home/matlab/python1.py =====
enter the number24
24 is not a prime
>>>
```

The status bar at the bottom right indicates "Ln: 11 Col: 0".

2.

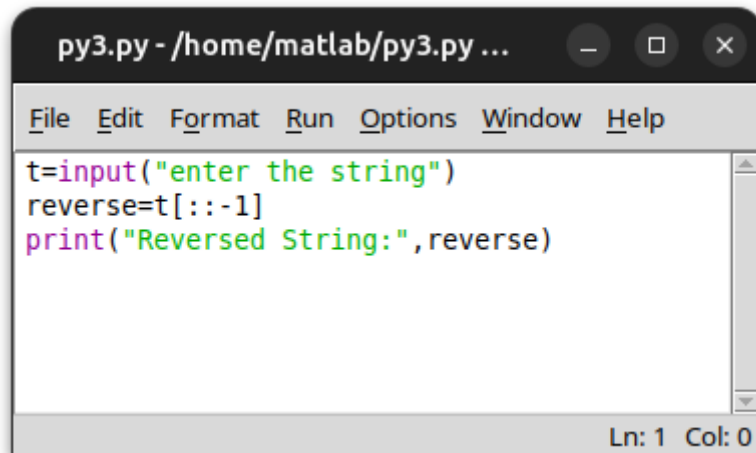


```
py2.py - /home/matlab/py2.py (3.12.3)
File Edit Format Run Options Window Help
n= int(input("enter the number of terms:"))
a,b=0,1
for i in range(n):
    print(a,end=" ")
    a,b=b,a+b
Ln: 1 Col: 0
```

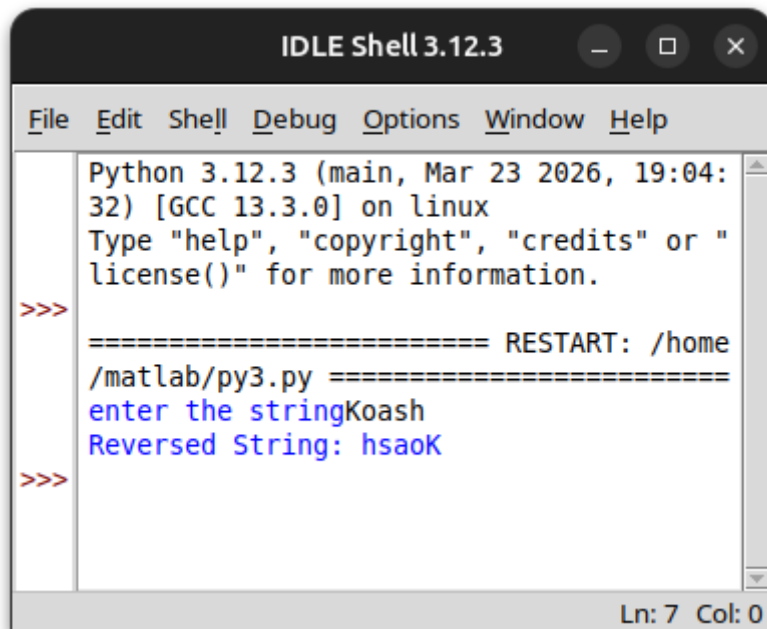


```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (main, Mar 23 2026, 19:04:32)
[GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: /home/matlab/py2.py =====
enter the number of terms:6
0 1 1 2 3 5
>>>
Ln: 7 Col: 0
```

3.

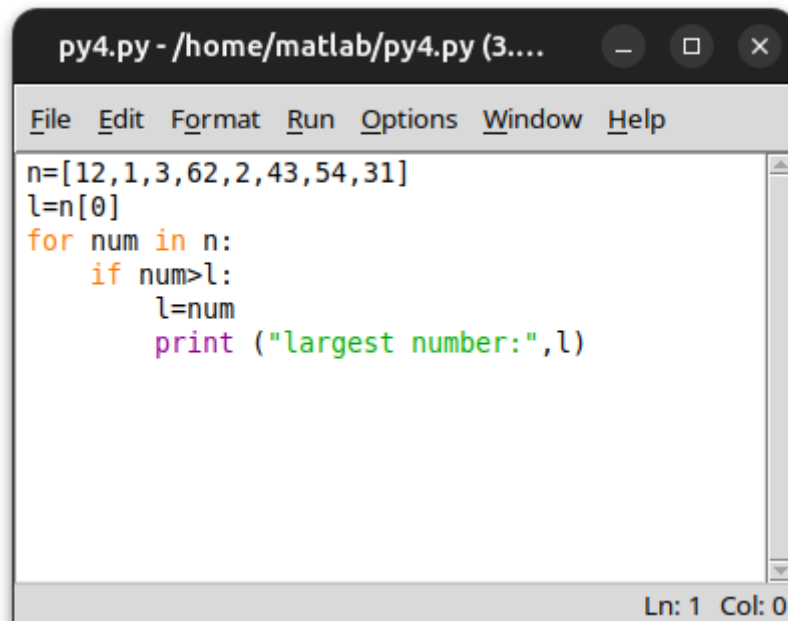


```
py3.py - /home/matlab/py3.py ...
File Edit Format Run Options Window Help
t=input("enter the string")
reverse=t[::-1]
print("Reversed String:",reverse)
Ln: 1 Col: 0
```

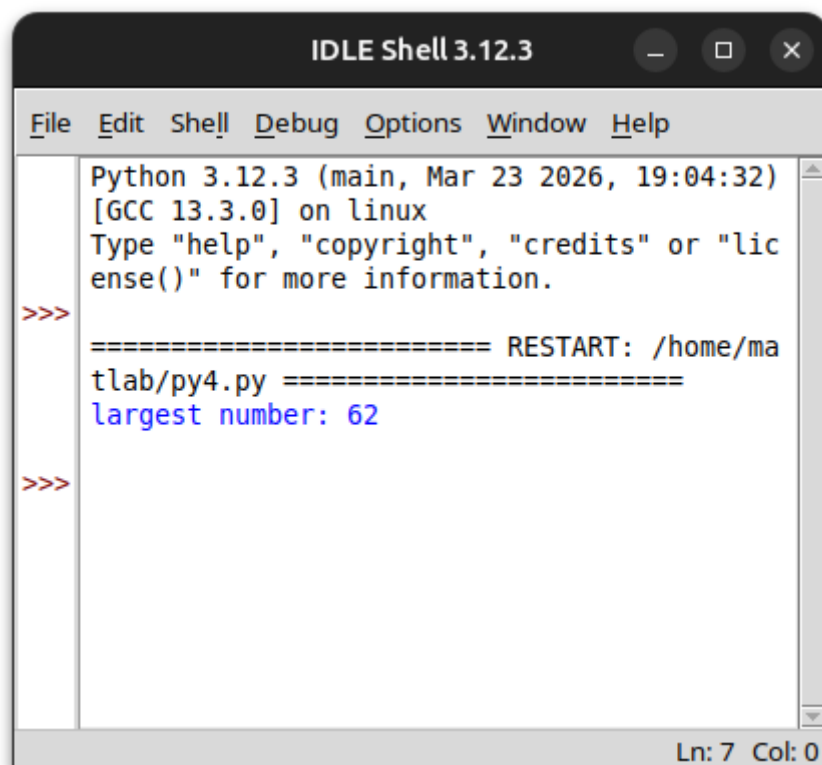


```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: /home/matlab/py3.py =====
enter the stringKoash
Reversed String: hsaok
>>>
Ln: 7 Col: 0
```

4.

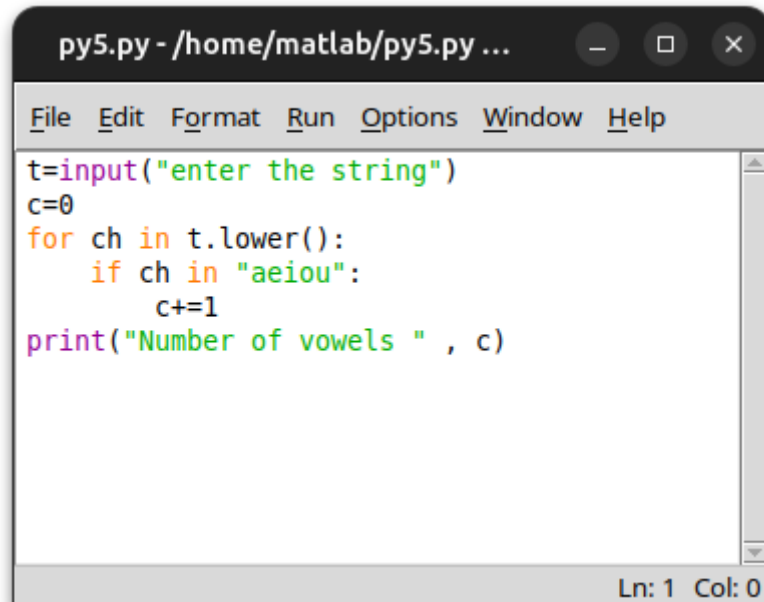


```
py4.py - /home/matlab/py4.py (3...  
File Edit Format Run Options Window Help  
n=[12,1,3,62,2,43,54,31]  
l=n[0]  
for num in n:  
    if num>l:  
        l=num  
    print ("largest number:",l)  
Ln: 1 Col: 0
```

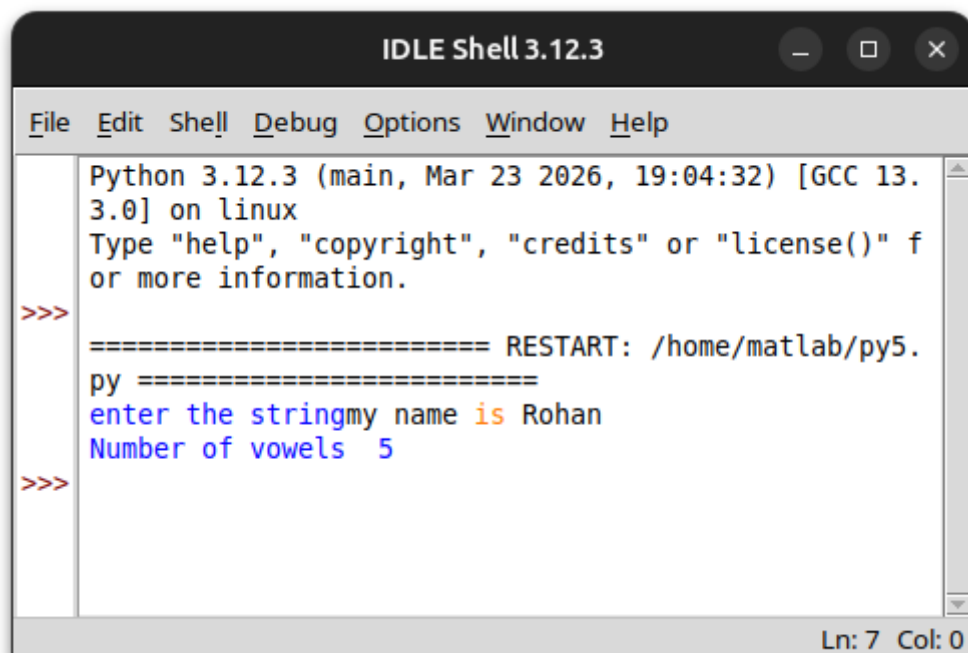


```
IDLE Shell 3.12.3  
File Edit Shell Debug Options Window Help  
Python 3.12.3 (main, Mar 23 2026, 19:04:32)  
[GCC 13.3.0] on linux  
Type "help", "copyright", "credits" or "lic  
ense()" for more information.  
>>> ===== RESTART: /home/ma  
tlab/py4.py =====  
largest number: 62  
>>>  
Ln: 7 Col: 0
```

5.

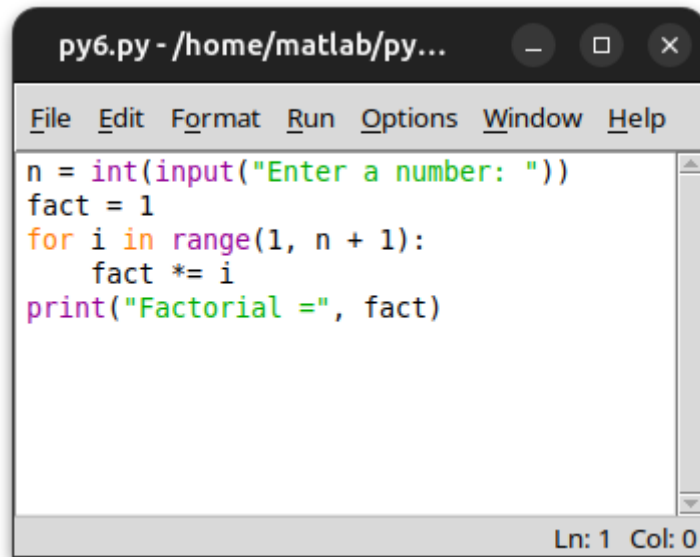


```
py5.py - /home/matlab/py5.py ...
File Edit Format Run Options Window Help
t=input("enter the string")
c=0
for ch in t.lower():
    if ch in "aeiou":
        c+=1
print("Number of vowels " , c)
Ln: 1 Col: 0
```

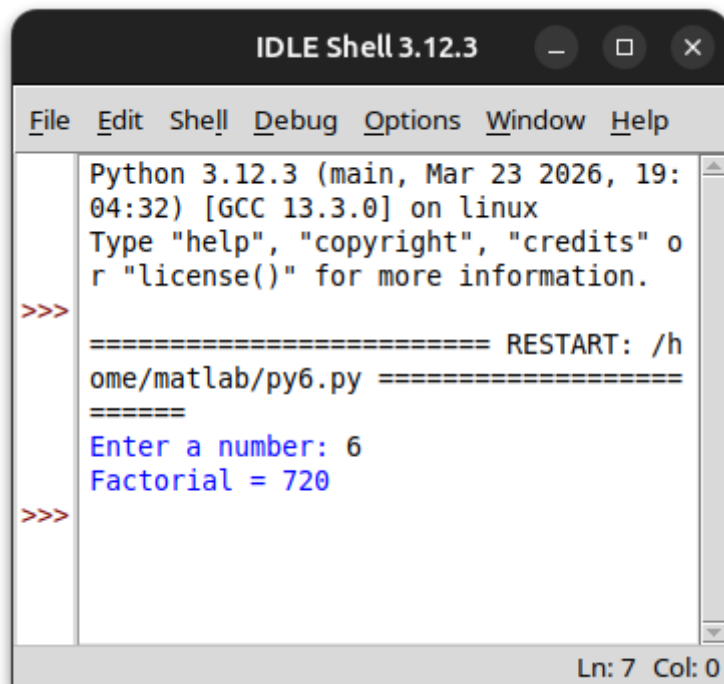


```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: /home/matlab/py5.py =====
>>> enter the string
my name is Rohan
Number of vowels 5
>>>
Ln: 7 Col: 0
```

6.

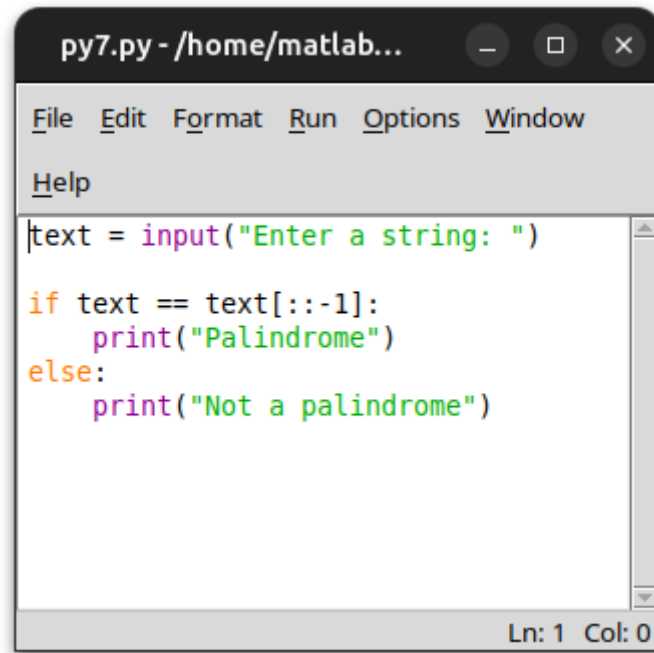


```
py6.py - /home/matlab/py...
File Edit Format Run Options Window Help
n = int(input("Enter a number: "))
fact = 1
for i in range(1, n + 1):
    fact *= i
print("Factorial =", fact)
Ln: 1 Col: 0
```

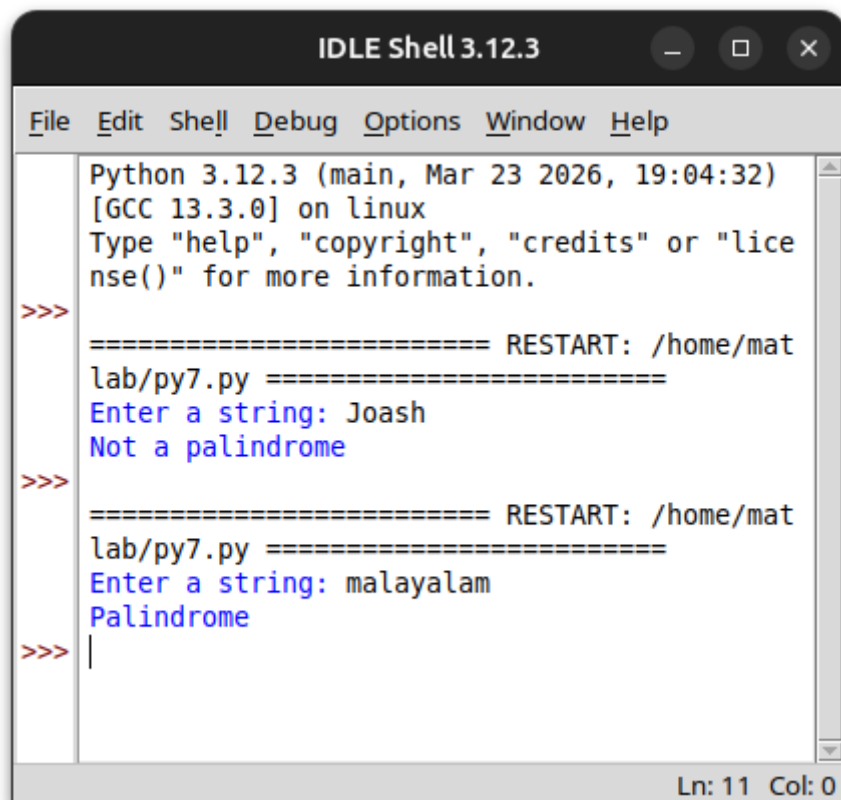


```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: /home/matlab/py6.py =====
>>> Enter a number: 6
Factorial = 720
>>>
Ln: 7 Col: 0
```

7.

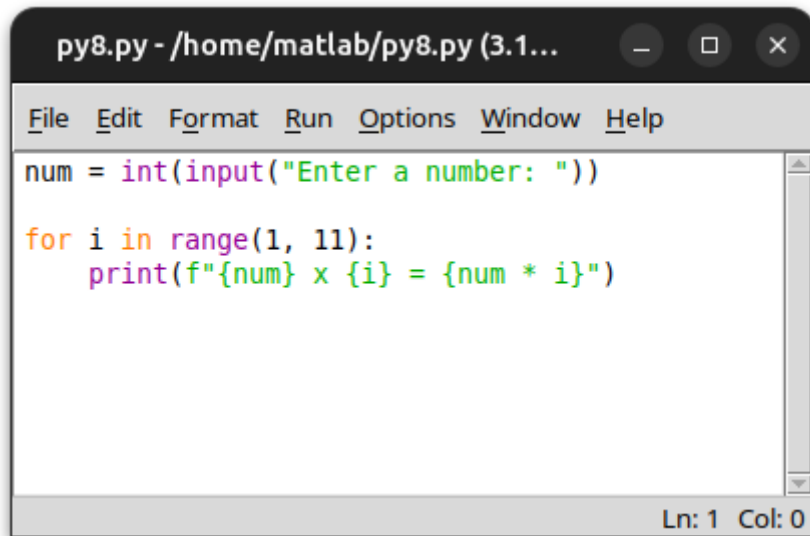


```
py7.py - /home/matlab...
File Edit Format Run Options Window
Help
text = input("Enter a string: ")
if text == text[::-1]:
    print("Palindrome")
else:
    print("Not a palindrome")
Ln: 1 Col: 0
```

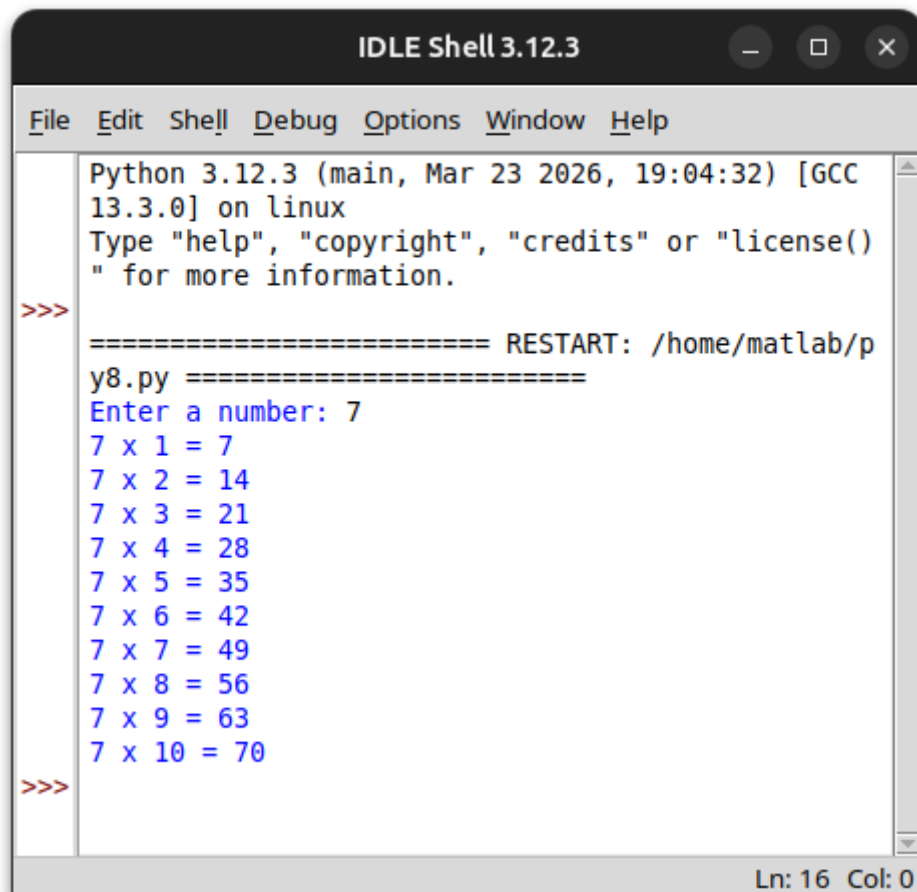


```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (main, Mar 23 2026, 19:04:32)
[GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: /home/matlab/py7.py =====
Enter a string: Joash
Not a palindrome
>>>
===== RESTART: /home/matlab/py7.py =====
Enter a string: malayalam
Palindrome
>>>
Ln: 11 Col: 0
```


8.

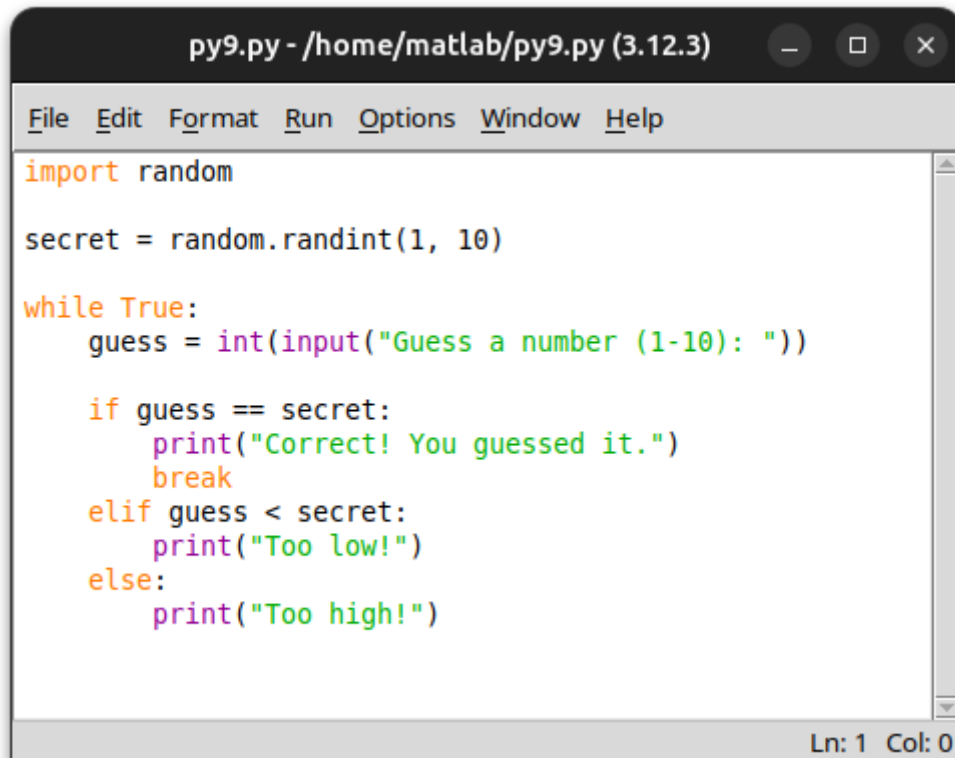


```
py8.py - /home/matlab/py8.py (3.1...  
File Edit Format Run Options Window Help  
num = int(input("Enter a number: "))  
  
for i in range(1, 11):  
    print(f"{num} x {i} = {num * i}")  
  
Ln: 1 Col: 0
```



```
IDLE Shell 3.12.3  
File Edit Shell Debug Options Window Help  
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC  
13.3.0] on linux  
Type "help", "copyright", "credits" or "license()  
" for more information.  
>>> ===== RESTART: /home/matlab/p  
y8.py =====  
Enter a number: 7  
7 x 1 = 7  
7 x 2 = 14  
7 x 3 = 21  
7 x 4 = 28  
7 x 5 = 35  
7 x 6 = 42  
7 x 7 = 49  
7 x 8 = 56  
7 x 9 = 63  
7 x 10 = 70  
>>>  
  
Ln: 16 Col: 0
```

9.



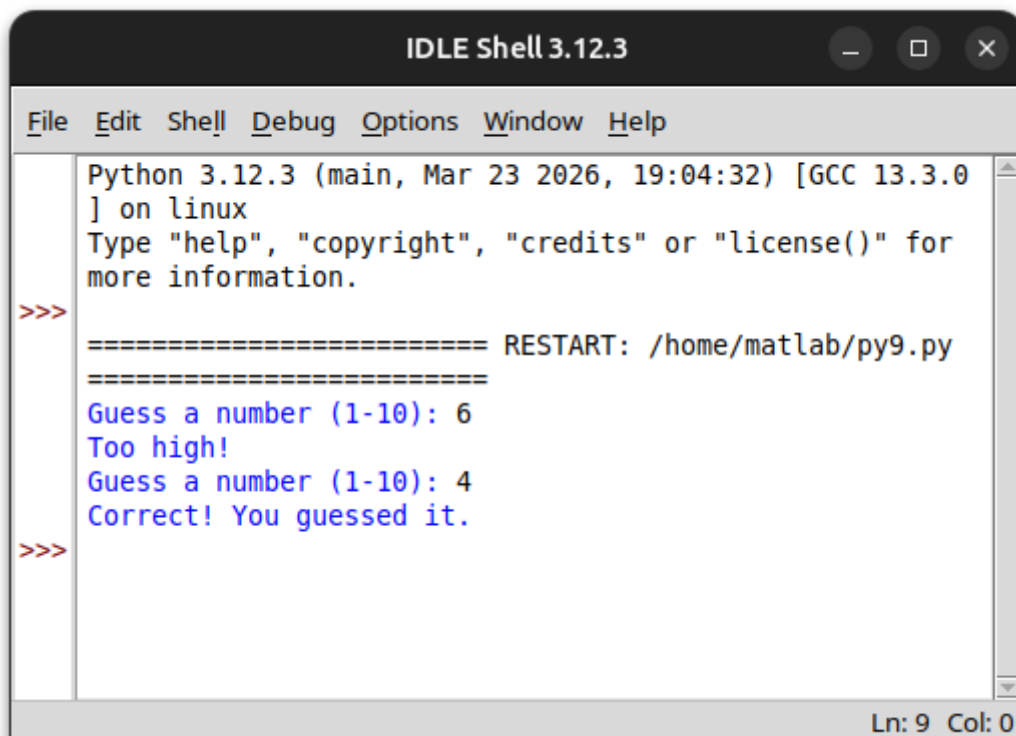
```
py9.py - /home/matlab/py9.py (3.12.3)
File Edit Format Run Options Window Help
import random

secret = random.randint(1, 10)

while True:
    guess = int(input("Guess a number (1-10): "))

    if guess == secret:
        print("Correct! You guessed it.")
        break
    elif guess < secret:
        print("Too low!")
    else:
        print("Too high!")

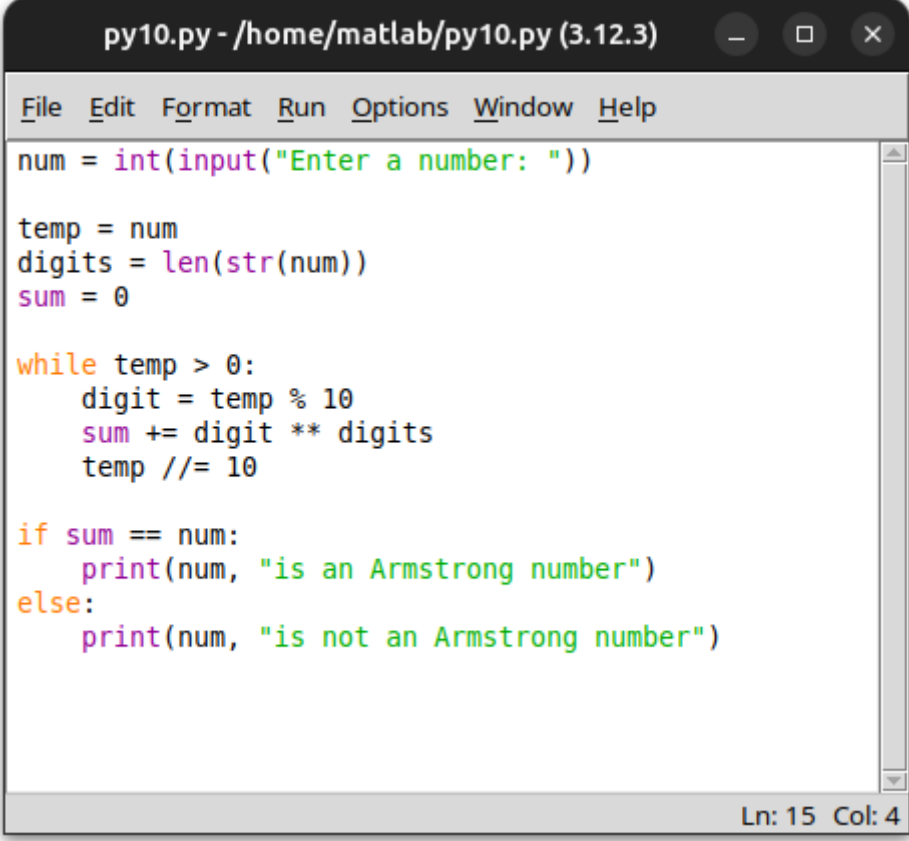
Ln: 1 Col: 0
```



```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0
] on linux
Type "help", "copyright", "credits" or "license()" for
more information.
>>>
===== RESTART: /home/matlab/py9.py
=====
Guess a number (1-10): 6
Too high!
Guess a number (1-10): 4
Correct! You guessed it.
>>>

Ln: 9 Col: 0
```

10.



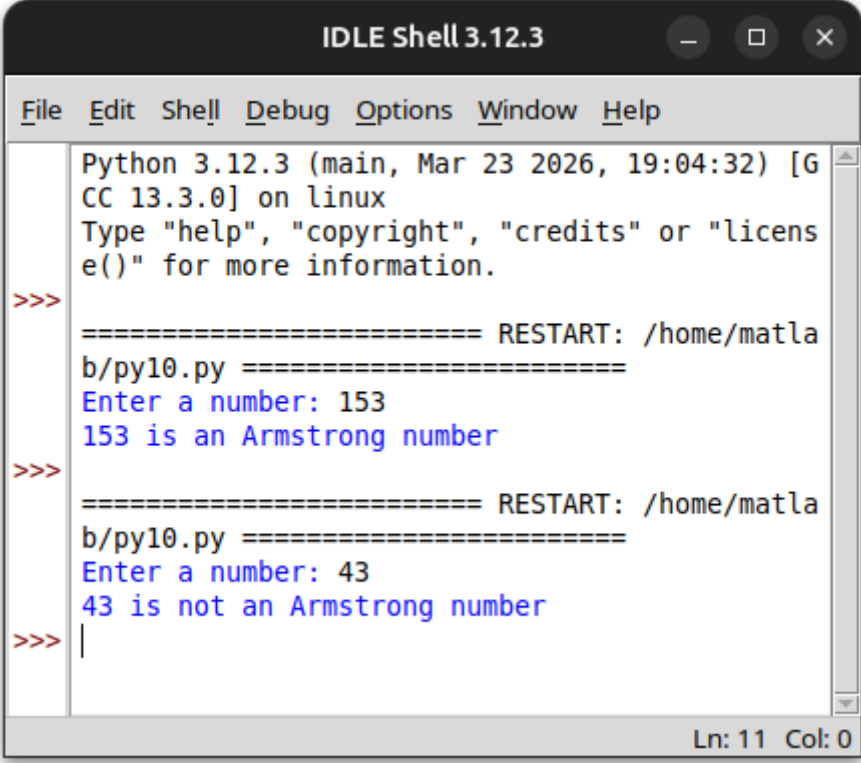
```
py10.py - /home/matlab/py10.py (3.12.3)
File Edit Format Run Options Window Help
num = int(input("Enter a number: "))

temp = num
digits = len(str(num))
sum = 0

while temp > 0:
    digit = temp % 10
    sum += digit ** digits
    temp //= 10

if sum == num:
    print(num, "is an Armstrong number")
else:
    print(num, "is not an Armstrong number")

Ln: 15 Col: 4
```



```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>> ===== RESTART: /home/matlab/py10.py =====
Enter a number: 153
153 is an Armstrong number
>>> ===== RESTART: /home/matlab/py10.py =====
Enter a number: 43
43 is not an Armstrong number
>>> |

Ln: 11 Col: 0
```